

1865 ▶ 1875

1867

Dynamite!

Alfred Nobel, a Swedish chemist, invented an explosive powder that he called "dynamite." Due to dynamite later being used on the battlefield, Nobel realized his legacy had become associated with death. To change this, in his will Nobel left a large amount of his money to fund the awards that became known as the Nobel Prizes—including one dedicated to those who promote peace.

1867

Power on two wheels

Three years after French blacksmith Pierre Michaux built the first iron-framed pedal bicycle, Louis-Guillaume Perreaux attached a small alcohol-fueled steam engine to it. Their machine is often said to be the world's first motorcycle. Only one model was ever built.

Pressurized steam enters the cylinder when the bike starts to move.

Leather drive belts transfer power to the back wheel

Model of the Michaux-Perreaux steam motorcycle

Foot pedals set the cycle in motion



1865

1865

Safer surgery

Scottish surgeon Joseph Lister realized patients were dying because their wounds became infected with bacteria. He introduced carbolic acid as an antiseptic to kill bacteria in the operating room. It worked and death rates fell, but carbolic acid proved harmful both to patient and surgeon, and Lister later replaced it with boric acid.

1868

Helium discovered

French astronomer Jules Janssen was watching a total eclipse of the Sun when he spotted a line of yellow light in the Sun's spectrum that did not match the wavelength of any known element. He had discovered helium—the second lightest and second most abundant element in the Universe, after hydrogen.

1869

DNA discovery

Swiss physician Friedrich Miescher discovered a substance inside the nuclei of cells. He called it "nuclein." We now know it to be DNA, the material that carries the genetic instructions for all living organisms.

Early modern human

A skull discovered in southwestern France was identified as that of an early modern human (Cro-Magnon man), the first example found in Western Europe.

The code of life
See pages 198–199

